

Quarto Proto Doc

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2026-09-03

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1 Introduction

Most statistical models can be seen as a special case or extension of the General Linear Model:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{1}$$

For Eqn. 1, A-optimality of a design is based on $\text{tr}((\mathbf{X}^\top \mathbf{X})^{-1})$.

Pearson ([1905](#)) is an old reference, but there are older ones ([Spearman 1904](#)).

2 References

Pearson, Karl. 1905. “On the Inheritance of Insanity.” *British Medical Journal* 1 (2317): 1175–77. <https://doi.org/10.1136/bmj.1.2317.1175-a>.

Spearman, C. 1904. “The Proof and Measurement of Association Between Two Things.” *The American Journal of Psychology* 15 (1): 72. <https://doi.org/10.2307/1412159>.